

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

2018 HSC Mathematics Extension 1 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	A
3	A
4	D
5	A
6	C
7	C
8	B
9	D
10	B

Section II

Question 11 (a) (i)

Criteria	Marks
• Provides correct proof	

Sample answer:

$$\begin{aligned}
 P(1) &= 1^3 - 2(1)^2 - 5(1) + 6 \\
 &= 1 - 2 - 5 + 6 \\
 &= 0
 \end{aligned}$$

$\therefore x = 1$ is a zero of $P(x)$

Question 11 (a) (ii)

Criteria	Marks
• Finds correct solution	2
• Correctly divides by $x - 1$, or equivalent merit	1

Sample answer:

$$\begin{array}{r}
 x^2 - x - 6 \\
 x - 1 \overline{) x^3 - 2x^2 - 5x + 6} \\
 \underline{x^3 - x^2} \\
 -x^2 - 5x \\
 \underline{-x^2 + x} \\
 -6x + 6 \\
 \underline{-6x + 6} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore P(x) &= (x - 1)(x^2 - x - 6) \\
 &= (x - 1)(x + 2)(x - 3)
 \end{aligned}$$

other zeros are $-2, 3$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Obtains correct linear equation, or equivalent merit	1

Sample answer:

$$\log_2 5 + \log_2 (x - 2) = 3$$

$$\log_2 (5(x - 2)) = 3$$

$$5(x - 2) = 2^3$$

$$5x - 10 = 8$$

$$5x = 18$$

$$x = \frac{18}{5} \text{ or } 3\frac{3}{5}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Correctly finds R , or equivalent merit	1

Sample answer:

$$\sqrt{3} \sin x + \cos x = R \sin(x + \alpha), \quad R > 0, \quad 0 \leq \alpha \leq \frac{\pi}{2}$$

$$= R \sin x \cos \alpha + R \cos x \sin \alpha$$

$$\therefore R \cos \alpha = \sqrt{3} \quad \textcircled{1}$$

$$R \sin \alpha = 1 \quad \textcircled{2}$$

$$\textcircled{2} \div \textcircled{1} \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\textcircled{2}^2 + \textcircled{1}^2 \quad R^2 (\sin^2 \alpha + \cos^2 \alpha) = 1 + 3$$

$$R^2 = 4$$

$$R = 2 \quad (R > 0)$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin \left(x + \frac{\pi}{6} \right)$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Attempts to use secant theorem resulting in a quadratic equation, or equivalent merit	1

Sample answer:

$$x(x+2) = 3(3+5)$$

$$x^2 + 2x = 24$$

$$x^2 + 2x - 24 = 0$$

$$(x+6)(x-4) = 0$$

$$x = -6 \text{ or } 4$$

But $x > 0 \therefore x = 4$

Question 11 (e) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Domain $4x - 1 \neq 0$

$$4x \neq 1$$

$$x \neq \frac{1}{4}$$

Domain: all real $x \neq \frac{1}{4}$

Question 11 (e) (ii)

Criteria	Marks
• Provides correct solution	2
• Successfully deals with the denominator, or equivalent merit	1

Sample answer:

$$f(x) < 1$$

$$\frac{1}{4x-1} < 1 \quad x \neq \frac{1}{4}$$

$$(4x-1)^2 \times \frac{1}{4x-1} < (4x-1)^2 \times 1$$

$$4x-1 < (4x-1)^2$$

$$0 < (4x-1)^2 - (4x-1)$$

$$0 < (4x-1)(4x-1-1)$$

$$0 < (4x-1)(4x-2)$$

$$0 < 2(4x-1)(2x-1)$$

$$\therefore x < \frac{1}{4}, x > \frac{1}{2}$$

Question 11 (f)

Criteria	Marks
• Provides correct solution	3
• Obtains correct integral in terms of u , including the limits, or equivalent merit	2
• Attempts to use the substitution, or equivalent merit	1

Sample answer:

$$I = \int_{-3}^0 \frac{x}{\sqrt{1-x}} dx$$

$$= \int_4^1 \frac{1-u}{\sqrt{u}} (-du)$$

$$= \int_1^4 u^{-\frac{1}{2}} - u^{\frac{1}{2}} du$$

$$= \left[2u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^4$$

$$= 2(2) - \frac{2}{3}(8) - \left(2(1) - \frac{2}{3}(1) \right)$$

$$= 4 - 5\frac{1}{3} - 1\frac{1}{3}$$

$$= -2\frac{2}{3}$$

$$\text{Let } u = 1 - x$$

$$du = -dx$$

$$x = -3 \rightarrow u = 4$$

$$x = 0 \rightarrow u = 1$$

Question 12 (a)

Criteria	Marks
• Provides correct solution	2
• Rewrites the integrand using double angle result, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \int \cos^2(3x) dx \\
 &= \frac{1}{2} \int \cos(6x) + 1 dx \quad \text{since } \cos^2 \theta = \frac{1}{2}(\cos 2\theta + 1) \\
 &= \frac{1}{2} \left(\frac{1}{6} \sin 6x + x \right) + c \\
 &= \frac{1}{12} \sin 6x + \frac{x}{2} + c
 \end{aligned}$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$\begin{aligned}
 \frac{h}{20} &= \sin \theta \\
 h &= 20 \sin \theta \\
 \frac{dh}{d\theta} &= 20 \cos \theta
 \end{aligned}$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses related rates to find an expression for $\frac{dh}{dt}$, or equivalent merit	1

Sample answer:

When $h = 15$ $\theta = ?$

$$\sin \theta = \frac{15}{20} = \frac{3}{4}$$

$$\theta = \sin^{-1}\left(\frac{3}{4}\right)$$

we want $\frac{dh}{dt} = \frac{dh}{d\theta} \times \frac{d\theta}{dt}$

$$\frac{dh}{d\theta} \text{ when } h = 15 \quad \frac{dh}{d\theta} = 20 \cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right)$$

and $\frac{d\theta}{dt} = 1.5$ radians/minute (given)

$$\begin{aligned} \text{so } \frac{dh}{dt} &= 20 \cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right) \times 1.5 \\ &= 30 \cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right) \\ &= 19.8 \end{aligned}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$f(x) = \sin^{-1} x + \cos^{-1} x$$

$$\begin{aligned} f'(x) &= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \\ &= 0 \end{aligned}$$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$f(x) = \sin^{-1} x + \cos^{-1} x$$

Since $f'(x) = 0$

$f(x)$ must have a constant value on its domain, so only need to find $f(a)$ for some a .

When $x = 0$

$$f(0) = \sin^{-1} 0 + \cos^{-1} 0$$

$$= 0 + \frac{\pi}{2} = \frac{\pi}{2}$$

or

$$\text{Let } m = \sin^{-1} x$$

$$\sin m = x$$

$$\therefore \cos\left(\frac{\pi}{2} - m\right) = x$$

$$\text{Let } \frac{\pi}{2} - m = \cos^{-1} x$$

$$\begin{aligned} \therefore \sin^{-1} x + \cos^{-1} x &= m + \frac{\pi}{2} - m \\ &= \frac{\pi}{2} \end{aligned}$$

Question 12 (c) (iii)

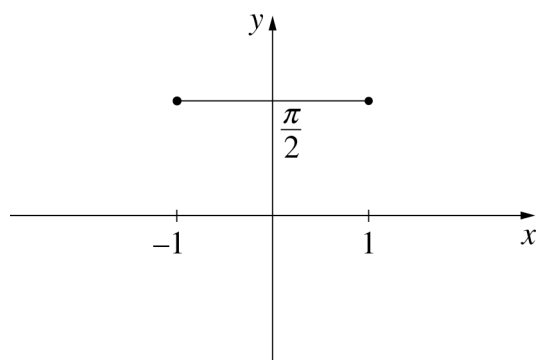
Criteria	Marks
• Provides correct sketch	1

Sample answer:

Sketch $f(x) = \sin^{-1} x + \cos^{-1} x$

domain $\sin^{-1} x$ is $-1 \leq x \leq 1$
 $\cos^{-1} x$ is $-1 \leq x \leq 1$

so domain $f(x)$ is $-1 \leq x \leq 1$



Question 12 (d)

Criteria	Marks
• Provides correct solution	2
• Obtains correct probability for 10 people completing the trek, or equivalent merit	1

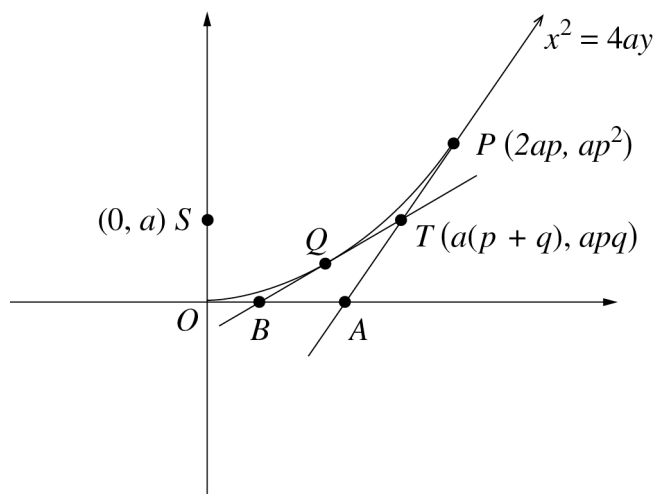
Sample answer:

$$\binom{12}{10}(0.75)^{10}(0.25)^2 + \binom{12}{11}(0.75)^{11}(0.25) + \binom{12}{12}(0.75)^{12}$$

Question 12 (e) (i)

Criteria	Marks
• Provides correct solution	2
• Finds the gradient of AS , or equivalent merit	1

Sample answer:



Tangent at P $y = px - ap^2$

so the coordinates of A are $(ap, 0)$

$$\begin{aligned}
 \text{the line joining } A \text{ and } S \text{ has slope} &= \frac{a-0}{0-ap} \\
 &= -\frac{a}{ap} \\
 &= -\frac{1}{p}
 \end{aligned}$$

$$\begin{aligned}
 \text{slope of tangent} \times \text{slope of } AS &= -\frac{1}{p} \times p \\
 &= -1
 \end{aligned}$$

lines are perpendicular

so $\angle PAS = 90^\circ$

Question 12 (e) (ii)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\angle PAS = 90^\circ \text{ which means } \angle TAS = 90^\circ$$

similarly we could show that

$$\angle QBS = 90^\circ \text{ so } \angle TBS = 90^\circ$$

So we have equal angles subtended on the same chord (on the same arc).

So the points A , B , S and T are concyclic points.

Question 12 (e) (iii)

Criteria	Marks
• Provides correct solution	2
• Correctly substitutes S and T into the distance formula, or equivalent merit	1

Sample answer:

ST is the diameter of the circle since $\angle TAS = \angle TBS = 90^\circ$

so we want the length of interval ST

coordinates of $S = (0, a)$ coordinates $T = (a(p + q), apq)$

$$\begin{aligned}
 \text{distance} &= \sqrt{(a(p + q) - 0)^2 + (apq - a)^2} \\
 &= \sqrt{a^2(p + q)^2 + a^2(pq - 1)^2} \\
 &= \sqrt{a^2(p^2 + 2pq + q^2 + p^2q^2 - 2pq + 1)} \\
 &= a\sqrt{p^2 + q^2 + 1 + p^2q^2} \\
 &= a\sqrt{(p^2 + 1)(q^2 + 1)}
 \end{aligned}$$

Question 13 (a)

Criteria	Marks
• Provides correct proof	3
• Correctly sets up S_{k+1} as the sum of S_k and T_{k+1} , or equivalent merit	2
• Proves true for $n = 1$, or equivalent merit	1

Sample answer:

$$\text{RTP: } 2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}$$

Show true for $n = 1$:

$$\begin{aligned} \text{LHS} &= 2(-3)^{1-1} \\ &= 2 \times 1 \\ &= 2 \end{aligned} \qquad \begin{aligned} \text{RHS} &= \frac{1 - (-3)^1}{2} \\ &= \frac{1 + 3}{2} \\ &= 2 \end{aligned}$$

LHS = RHS $\therefore$ true for $n = 1$

Assume true for $n = k$:

$$2 - 6 + 18 - 54 + \dots + 2(-3)^{k-1} = \frac{1 - (-3)^k}{2}$$

If $n = k$ is true, show true for $n = k + 1$.

$$\begin{aligned} \text{LHS} &= 2 - 6 + 18 - 54 + \dots + 2(-3)^{k-1} + 2(-3)^k \\ &= \frac{1 - (-3)^k}{2} + 2(-3)^k \\ &= \frac{1 - (-3)^k + 4(-3)^k}{2} \\ &= \frac{1 - (-3)(-3)^k}{2} \\ &= \frac{1 - (-3)^{k+1}}{2} \end{aligned}$$

$\therefore$ true for $n = k + 1$.

Hence, by the process of mathematical induction $2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}$.

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• States correct domain, or equivalent merit	1

Sample answer:

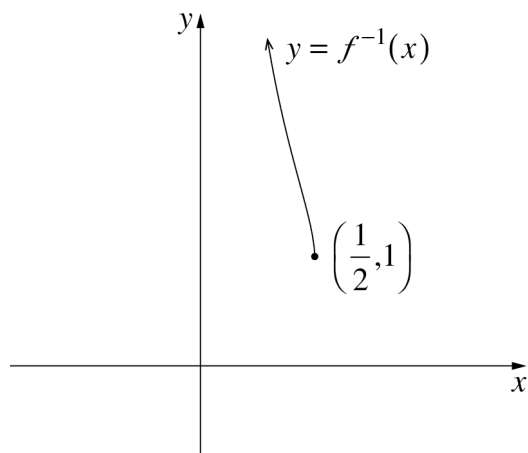
Domain: all real x , $0 < x \leq \frac{1}{2}$

Range: all real, $y \geq 1$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct sketch	1

Sample answer:



Question 13 (b) (iii)

Criteria	Marks
• Provides correct solution	3
• Correctly solves the resulting quadratic equation, or equivalent merit	2
• Correctly interchanges x and y , or equivalent merit	1

Sample answer:

Let $y = \frac{x}{x^2 + 1}$

$$x^2y + y = x$$

$$x^2y - x + y = 0$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times y \times y}}{2y}$$

$$= \frac{1 \pm \sqrt{1 - 4y^2}}{2y}$$

when $x = 2, y = \frac{2}{5}$

$$\therefore x = \frac{1 + \sqrt{1 - 4y^2}}{2y}$$

$$\therefore f^{-1}(x) = \frac{1 + \sqrt{1 - 4x^2}}{2x}$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to solve $y = 0$, or equivalent merit	1

Sample answer:

The object hits the ground, after $t = 0$ when

$$\begin{aligned}
 0 &= y(t) \\
 &= Vt \sin \theta - \frac{gt^2}{2} \\
 &= t \left(V \sin \theta - \frac{gt}{2} \right)
 \end{aligned}$$

As $t > 0$ we have

$$\begin{aligned}
 t &= \frac{2V \sin \theta}{g} \\
 x &= Vt \cos \theta \\
 &= \frac{2V^2 \sin \theta \cos \theta}{g} \\
 &= \frac{V^2}{g} \sin 2\theta
 \end{aligned}$$

Thus the range, at angle θ , is $\frac{V^2}{g} \sin 2\theta$.

Question 13 (c) (ii)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$\begin{aligned}
 \text{Horizontal range} &= \frac{V^2}{g} \sin 2 \left(\frac{\pi}{2} - \theta \right) \\
 &= \frac{V^2}{g} \sin(\pi - 2\theta) \\
 &= \frac{V^2}{g} \sin 2\theta
 \end{aligned}$$

Question 13 (c) (iii)

Criteria	Marks
• Provides correct solution	3
• Finds h_{α} or equivalent merit	2
• Finds time when maximum height is reached, or equivalent merit	1

Sample answer:

The maximum height, for angle θ , occurs when $\dot{y}(t) = 0$, that is

$$\begin{aligned}
 0 &= \dot{y}(t) \\
 &= V \sin \theta - gt \\
 t &= \frac{V \sin \theta}{g} \\
 y\left(\frac{V \sin \theta}{g}\right) &= \frac{V^2 \sin^2 \theta}{g} - \frac{V^2 \sin^2 \theta}{2g} \\
 &= \frac{V^2 \sin^2 \theta}{2g}
 \end{aligned}$$

Thus $h_{\alpha} = \frac{V^2 \sin^2 \alpha}{2g}$ and $h_{\beta} = \frac{V^2 \sin^2 \beta}{2g}$, but

$$\sin \beta = \sin\left(\frac{\pi}{2} - \alpha\right) = \cos \alpha$$

Thus

$$\begin{aligned}
 h_{\alpha} + h_{\beta} &= \frac{V^2 \sin^2 \alpha}{2g} + \frac{V^2 \sin^2 \beta}{2g} \\
 &= \frac{V^2 \sin^2 \alpha}{2g} + \frac{V^2 \cos^2 \alpha}{2g} \\
 &= \frac{V^2 (\sin^2 \alpha + \cos^2 \alpha)}{2g} \\
 &= \frac{V^2}{2g} \\
 \frac{h_{\alpha} + h_{\beta}}{2} &= \frac{V^2}{4g}
 \end{aligned}$$

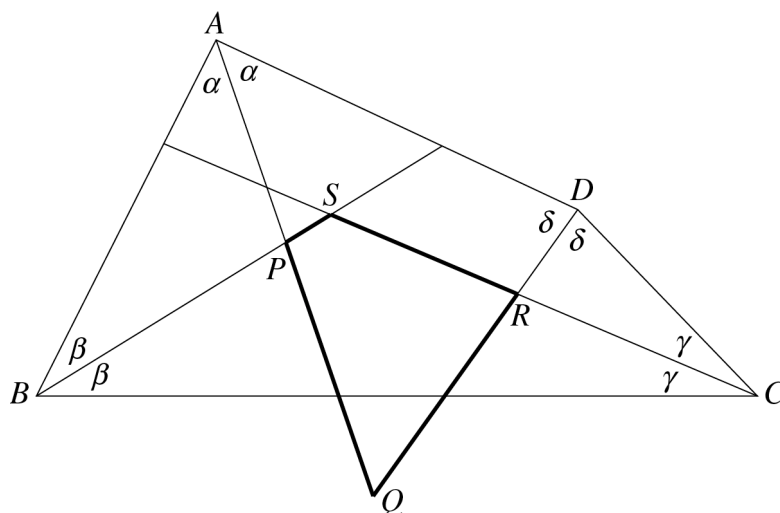
which depends only on V and g .

In fact, this is the height attained by the object when projected at $\frac{\pi}{4}$ to attain its maximum range.

Question 14 (a)

Criteria	Marks
• Provides correct proof	3
• Establishes a relationship between the angles of the quadrilateral $PQRS$ and the bisected angles, or equivalent merit	2
• Relates the sum of the bisected angles to the angle sum of the quadrilateral $ABCD$, or equivalent merit	1

Sample answer:



$$\text{Let } \alpha = \frac{1}{2}\angle A, \quad \beta = \frac{1}{2}\angle B$$

$$\gamma = \frac{1}{2}\angle C \quad \text{and} \quad \delta = \frac{1}{2}\angle D$$

as shown

$$\text{Then } 2\alpha + 2\beta + 2\gamma + 2\delta = 360^\circ \quad (\angle \text{ sum quadrilateral})$$

$$\therefore \alpha + \beta + \gamma + \delta = 180^\circ$$

$$\angle PSR = \angle BSC = 180^\circ - (\beta + \gamma) \quad (\angle \text{ sum of } \triangle BSC)$$

$$\angle PQR = \angle AQD = 180^\circ - (\alpha + \delta) \quad (\angle \text{ sum of } \triangle AQD)$$

$$\begin{aligned} \therefore \angle PSR + \angle PQR &= 2 \times 180^\circ - (\alpha + \beta + \gamma + \delta) \\ &= 2 \times 180^\circ - 180^\circ \\ &= 180^\circ \end{aligned}$$

$\therefore PQRS$ is cyclic (opposite angles are supplementary)

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Obtains correct expansion of $(1 + (1 + x)^n)$ in terms of $(1 + x)^k$, or equivalent merit	2
• Correctly applies the binomial theorem to one binomial expression to get the coefficient of x^r , or equivalent merit	1

Sample answer:

Consider the coefficients of x^r on both sides of the equation

$$(2 + x)^n = (1 + (1 + x))^n$$

$$\text{Then in } x^r \text{ on LHS} = \binom{n}{r} x^r 2^{n-r}$$

$$\therefore \text{coefficient of } x^r = \binom{n}{r} 2^{n-r}$$

$$\text{RHS} = \binom{n}{0} + \binom{n}{1}(1 + x)^1 + \binom{n}{2}(1 + x)^2 + \dots + \binom{n}{r}(1 + x)^r + \binom{n}{r+1}(1 + x)^{r+1} + \dots + \binom{n}{n}(1 + x)^n$$

Coefficient of x^r = sum of coefficients of x^r for each term

$$= \binom{n}{r} \binom{r}{r} + \binom{n}{r+1} \binom{r+1}{r} + \binom{n}{r+2} \binom{r+2}{r} + \dots + \binom{n}{n} \binom{n}{r}$$

Equating these gives the result.

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the formula in part (i), or equivalent merit	1

Sample answer:

Selector *A* could choose a group of 4, 5, 6 ... or 23 people. Then selector *B* chooses 4 from this group.

Hence number of ways

$$\begin{aligned}
 &= \binom{23}{4} \binom{4}{4} + \binom{23}{5} \binom{5}{4} + \binom{23}{6} \binom{6}{4} + \dots + \binom{23}{23} \binom{23}{4} \\
 &= \binom{23}{4} 2^{23-4} \\
 &= \binom{23}{4} (2^{19}) \\
 &= 4\,642\,570\,240
 \end{aligned}$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

In $\triangle ABC$ and $\triangle ACD$

$\angle ACB = \angle ADC$ (CD perpendicular to AB , $\triangle ABC$ right-angled at C)

$\angle BAC = \angle CAD$ (common)

$\therefore \triangle ABC \parallel \triangle ACD$ (equiangular)

Question 14 (c) (ii)

Criteria	Marks
• Provides correct proof	1

Sample answer:

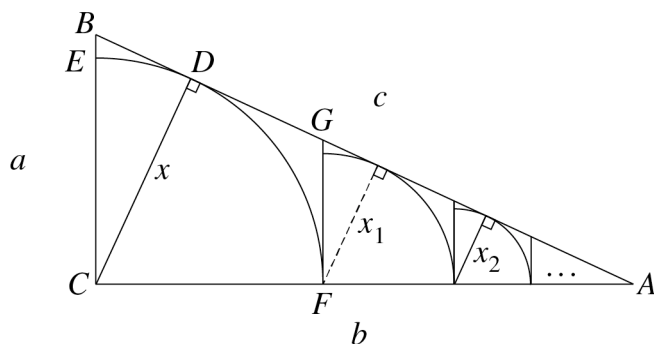
$$\frac{x}{a} = \frac{b}{c} \quad (\text{corresponding sides of similar triangles})$$

$$\therefore x = \frac{ab}{c}$$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct solution	4
• Correctly establishes a relationship between successive radii or area, or equivalent merit	3
• Attempts to use similarity to find a relationship between successive radii or area, or equivalent merit	2
• Finds an expression for the area of a quadrant in terms of a , b and c , or equivalent merit	1

Sample answer:



Since $\triangle ABC$ and $\triangle AGF$ are right-angled and angle A is common, then $\triangle ABC \parallel \triangle AGF$

$$\frac{GF}{a} = \frac{b-x}{b}$$

$$\frac{GF}{a} = \frac{1}{b} \left(b - \frac{ab}{c} \right)$$

$$\frac{GF}{a} = \frac{b}{b} \left(1 - \frac{a}{c} \right)$$

It follows then

$$\frac{x_1}{x} = 1 - \frac{a}{c}$$

$$x_1 = \left(1 - \frac{a}{c} \right) x$$

$$x_1 = (1 - \sin A)x$$

The ratio of the lengths of the radii are independent of a , b and c .

Hence, successive radii will be in the ratio $(1 - \sin A)$.

$$\text{Hence, } x_2 = \left(1 - \frac{a}{c} \right)^2 x \text{ etc}$$

Area of quadrants

$$A = \frac{1}{4}\pi(x^2 + x_1^2 + x_2^2 + \dots)$$

$$A = \frac{1}{4}\pi\left(x^2 + \left(1 - \frac{a}{c}\right)^2 x^2 + \left(1 - \frac{a}{c}\right)^4 x^2 + \dots\right)$$

$$A = \frac{1}{4}\pi x^2 \left(1 + \left(1 - \frac{a}{c}\right)^2 + \left(1 - \frac{a}{c}\right)^4 + \dots\right)$$

For the series

$$1 + \left(1 - \frac{a}{c}\right)^2 + \left(1 - \frac{a}{c}\right)^4 + \dots$$

$$a = 1, \quad r = \left(1 - \frac{a}{c}\right)^2$$

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{1}{1 - \left(1 - \frac{a}{c}\right)^2}$$

$$S_{\infty} = \frac{c^2}{c^2 - (c-a)^2}$$

$$\begin{aligned} S_{\infty} &= \frac{c^2}{c^2 - c^2 + 2ac - a^2} \\ &= \frac{c^2}{a(2c-a)} \end{aligned}$$

$$\therefore A = \frac{1}{4}\pi x^2 \times \frac{c^2}{a(2c-a)}$$

$$A = \frac{\pi}{4} \left(\frac{ab}{c}\right)^2 \times \frac{c^2}{a(2c-a)}$$

$$A = \frac{\pi}{4} \times \frac{ab^2}{(2c-a)}$$

$$A = \frac{\pi ab^2}{4(2c-a)}$$

Question 14 (c) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Area of quadrants < area of $\triangle ABC$

$$\frac{\pi ab^2}{4(2c-a)} < \frac{ab}{2}$$

$$\frac{\pi}{4} < \frac{ab}{2} \times \frac{(2c-a)}{ab^2}$$

$$\frac{\pi}{4} < \frac{2c-a}{2b}$$

$$\frac{\pi}{2} < \frac{2c-a}{b}$$

2018 HSC Mathematics Extension 1 Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	16.3	PE3
2	1	6.6	PE6
3	1	5.7, 13.4	HE7
4	1	16.1	PE3
5	1	14.2E	HE3
6	1	16.4	HE7
7	1	14.3E	HE5
8	1	18.1	PE3
9	1	15.2	HE7
10	1	14.4E	HE5

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	1	16.2	PE3
11 (a) (ii)	2	16.2	PE3
11 (b)	2	12.2	H3, HE7
11 (c)	2	5.9E, 13.1	HE7
11 (d)	2	2.10	PE3
11 (e) (i)	1	4.1	PE2
11 (e) (ii)	2	1.4E	PE3
11 (f)	3	11.5E	HE6
12 (a)	2	13.6E	HE7
12 (b) (i)	1	14.1E, 14.4E	HE3, HE5
12 (b) (ii)	2	14.1E, 14.4E	HE3, HE5
12 (c) (i)	1	15.5	HE4
12 (c) (ii)	1	15.2, 15.4	HE4
12 (c) (iii)	1	15.3	HE4
12 (d)	2	18.2	HE3
12 (e) (i)	2	9.6	PE3
12 (e) (ii)	1	2.9	PE3
12 (e) (iii)	2	2.9, 6.5	PE3
13 (a)	3	7.4	HE2
13 (b) (i)	2	15.1	HE4
13 (b) (ii)	1	15.1E	HE4
13 (b) (iii)	3	15.1E	HE4
13 (c) (i)	2	14.3E	HE3
13 (c) (ii)	1	14.3E	HE3

Question	Marks	Content	Syllabus outcomes
13 (c) (iii)	3	14.3E	HE3
14 (a)	3	2.9E, 2.10E	PE3
14 (b) (i)	3	17.3	HE7
14 (b) (ii)	2	17.3	HE7
14 (c) (i)	1	2.3, 2.5	PE2, HE7
14 (c) (ii)	1	2.3, 2.5	PE2, HE7
14 (c) (iii)	4	7.3	PE2, PE6, HE7
14 (c) (iv)	1	1.4, 2.3, 2.5	PE2, PE6, HE7